

U.S. IT Hardware

Memory & Storage: What we learned from our NDR with STX Management and recent NAND & HDD earnings



Mark C. Newman
+1 212 845 7822
mark.newman@bernsteinsg.com



April Li
+1 917 344 8339
april.li@bernsteinsg.com



Phoebe Sun
+1 917 344 8481
phoebe.sun@bernsteinsg.com

Seagate has strong multi-year demand visibility, with expected volumes initially established through LTAs and then converted into firm PO (purchase orders) 4-5 quarters before shipment. Therefore, the June quarter's ~6% QoQ pricing increase primarily reflected pricing negotiated roughly four quarters earlier. Seagate also keeps a small amount of capacity uncommitted, allowing it to capture higher spot-market pricing on incremental volume. **Management expects HDD ASP growth to continue and potentially further accelerate as prior HAMR introductory discounts roll off, while edge and IoT products remain more flexible and can see faster price increases than nearline products.**

Industry HDD supply remains highly disciplined, as Seagate, WDC, and Toshiba are not adding meaningful unit capacity. Instead, Seagate expects to drive roughly 25% annual nearline exabyte growth through HAMR-based increases in capacity per drive, despite HAMR requiring more manufacturing resources and factory space. As the only vendor aggressively scaling HAMR, we believe STX is positioned to gain exabyte share while constrained industry supply supports favorable market dynamics.

Management sees HDD demand being driven by cloud, enterprise, AI training, and increasingly AI inference workloads. **The key takeaway is that AI creates more data, and most of that data ultimately ends up on low-cost HDD storage.** Longer data retention periods, combined with demand that is primarily for net-new capacity rather than refreshes, remain strong structural tailwinds for HDDs.

Management views neoclouds as a potential long-term HDD demand opportunity. While most neocloud providers currently rely on hyperscalers for storage, future expansion into full infrastructure platforms could require their own storage deployments, creating an additional source of HDD demand beyond traditional cloud customers.

Earnings takeaways: We believe the HDD pricing cycle is in its early innings with a long runway for continued ASP growth. Pricing only began inflecting in the March quarter and has been increasing steadily, supported by strong demand and disciplined industry supply. **STX is our top pick due to its HAMR leadership.** Its lead in areal density scaling should drive faster exabyte shipment growth, better cost reductions, and greater share gains versus peers. While **WDC** also posted a strong quarter and continues to benefit from improving industry pricing, its slower HAMR transition limits its cost decline relative to STX. **As a result, we remain constructive on WDC, but continue to view STX as the best-positioned beneficiary of the HDD upcycle. We also maintain Outperform rating on SNDK** and still see significant upside particularly as 2/3 of volumes on LTAs with 80% floor gross margin should help improve earnings stability.

STX is our top pick within the sector, due to continued HDD pricing strength, supply discipline, and HAMR leadership.

BERNSTEIN TICKER TABLE

Ticker	Rating	6 Aug 2026		Price Target	TTM Rel. Perf.	Adjusted EPS				Adjusted P/E (x)		
		Cur	Closing Price			Cur	2026A	2027E	2028E	2026A	2027E	2028E
STX (Seagate)	O	USD	852.95	1,350.00	457.7%	USD	15.53	40.61	64.40	54.9	21.0	13.2
WDC (Western Digital)	O	USD	451.52	770.00	490.5%	USD	10.19	23.39	36.58	44.3	19.3	12.3
SNDK (SanDisk)	O	USD	1,258.58	3,000.00	2868.0%	USD	70.08	238.36	272.09	18.0	5.3	4.6
SPX			7,709.96									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate SNDK Outperform with TP of \$3000. We rate STX Outperform with TP of \$1350. We rate WDC Outperform with TP of \$770.

DETAILS

Seagate uses two main forms of customer commitments: LTAs and POs. LTAs are multi-year agreements that primarily specify expected volumes, but generally do not lock in pricing or detailed product mix beyond the near-term period. **These LTA volumes are then converted into firm PO (purchase orders) approximately 4-5 quarters before shipment.** Each purchase order specifies volume, ASP, delivery timing, number of drives, and capacity per drive, and is relatively rigid once signed. Given approximately 3 quarters of production lead time, Seagate has enough time to set firm purchasing orders before committing manufacturing resources, providing strong visibility. The vast majority of Seagate's expected CY2027 exabyte shipments are already covered by purchase orders.

Nearline data center pricing is largely locked for the next 4-5 quarters through firm purchase orders, meaning reported pricing generally reflects negotiations completed several quarters earlier. Management indicated that the 6% qoq pricing increase realized in the June quarter was primarily based on pricing agreed roughly four quarters ago, with some additional benefit from market-priced incremental exabytes not under order. Seagate intentionally commits slightly less than its maximum production capability, allowing incremental in-quarter output to be sold at substantially higher market prices and providing a mechanism to test underlying demand. The pricing negotiations are staggered; therefore, STX enjoys ASP QoQ increases each quarter.

Non-nearline edge and IoT products generally do not have firm purchase orders or LTAs, allowing Seagate to raise prices more quickly. Data center and edge margins are currently not materially different, although data center is growing faster and edge demand is more seasonal as the consumer business is stronger in Sep-Dec quarters. Management expects steady HDD ASP growth to continue at a pace similar to, or potentially better than, recent quarters as prior introductory discounts on new HAMR products roll off.

None of STX, WDC, or Toshiba are adding drive unit capacity, keeping industry-wide HDD supply structurally disciplined. Seagate management explained that its factories are effectively full, and rather than expanding units, management is targeting roughly 25% CAGR in nearline exabyte shipments purely by increasing capacity per drive via HAMR. In fact, the HAMR transition itself requires longer production cycle times, more equipment, and additional factory space just to sustain current unit output, leaving little practical room for unit growth — and building a new factory would take at least two years, which management has no plans to do. Since STX is the only player seriously ramping HAMR, it also holds the best areal density potential in the industry, a scaling advantage neither WDC nor Toshiba can currently match. As a result, Seagate is directing available exabytes primarily to existing CSP customers and expects to grow exabyte share even with unit share unchanged, while the broader industry's inability to scale supply keeps supply disciplined and positions STX as the clear share gainer as the cycle progresses.

Management was unequivocal that HAMR remains the primary technology enabling future HDD capacity growth and economics. Rather than expanding drive unit production, Seagate intends to grow exabytes shipped through higher-capacity drives. The company argued that adding platters and heads offers diminishing returns, with moving from 10 to 11 disks increasing capacity by only ~10%, whereas migrating from 30TB to 40TB drives provides roughly a 30% capacity uplift. While HAMR requires more equipment, longer manufacturing cycle times, and additional factory space than ePMR, the company remains focused on technology transitions rather than factory expansion. Management stated they are effectively done adding manufacturing capacity and instead expect to achieve roughly 25% annual nearline exabyte growth through HAMR adoption, with the nearline portfolio already at approximately 40% HAMR mix and expected to become predominantly HAMR over time. Customer demand is increasingly concentrated on higher-capacity drives, and Seagate believes the current supply environment remains constrained despite factories operating at full utilization.

Management emphasized that HDD demand is not being driven by a single AI use case but rather by a combination of traditional cloud growth, enterprise storage, training workloads, and AI inference. The company believes it is increasingly difficult to separate what portion of demand is "AI" versus traditional storage demand because both ultimately result in growing data creation. The most important AI-related takeaway was that inference may become a more meaningful HDD demand driver than training over time. Every inference interaction generates additional context, outputs, metadata, and content that must be retained and stored, supporting longer-term demand for mass-capacity storage. Management also highlighted that data retention periods continue to lengthen, creating a structural tailwind for HDDs.

While NAND and SSDs are essential for high-performance workloads, KV cache, and warm storage, management noted that warm storage represents only a small portion of total exabytes, whereas cold storage remains significantly larger. The underlying message was that AI increases compute activity, compute creates data, and nearly all of that data ultimately finds its

way to lower-cost HDD storage. **Demand today is overwhelmingly for new capacity rather than refreshes**, as customers would rather add incremental drives than retire existing drives and lose revenue-generating storage capacity. HDDs are typically warranted for five years but often remain in service for six to seven years.

Neoclouds could become an incremental source of HDD demand over time. Today, most neocloud providers offer compute while relying on CSPs for storage, meaning Seagate's HDD shipments remain concentrated with hyperscalers. However, if neoclouds evolve into broader infrastructure platforms and seek greater independence from CSPs, they will likely need to build their own storage footprints, creating an additional layer of HDD demand. While this opportunity remains early, as neoclouds are not yet HAMR-qualified customers and pricing discussions have not begun, management views their emergence as a potential long-term growth vector beyond traditional cloud demand.

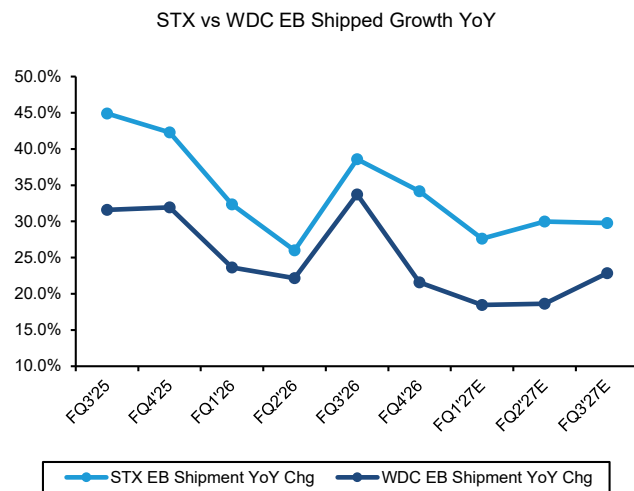
Earnings takeaways:

SNKD delivered a great FQ4'26. While some may worry its growing LTA coverage **caps some near-term upside** and is slowing ASP growth, **we believe it is important to note that greater LTA coverage is positive** for the stock's multiple and rerating potential: LTAs provide robust downside protection, **make nosebleed earnings more sustainable, and should reduce perceived cyclicality.** For additional details, please see [Sandisk FQ4'26: As LTAs hit 2/3 of volume, ASP increases slow but more downside protection than previously expected.](#)

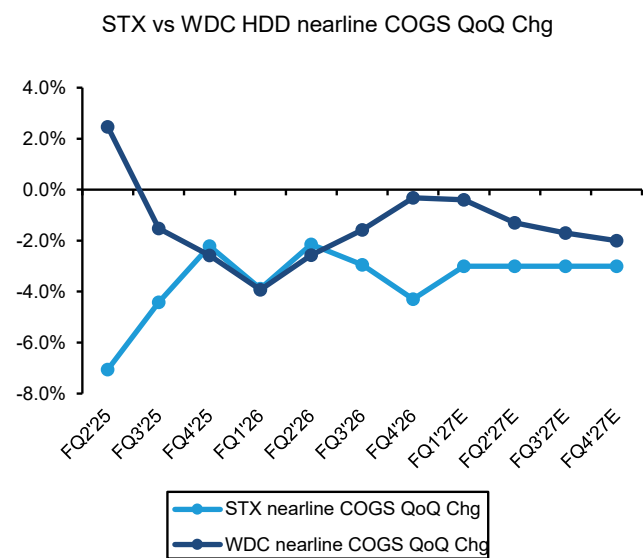
Why the HDD cycle has further to run: We believe the HDD ASP upcycle is just getting started. Pricing only inflected roughly two quarters ago and has since risen steadily in the mid-to-high single-digit range qoq— an early and steady ramp compared to memory's steep spike. Looking ahead, we would not be surprised if HDD qoq pricing growth actually steps up a bit from recent quarters, given strong end-market demand. Net, we see a long and steady runway ahead for HDD pricing.

STX vs. WDC: Within that favorable HDD backdrop, HAMR leadership is the key differentiator. Its lead in HAMR-driven areal density scaling supports faster exabyte shipment growth and deeper cost declines than peers. Moreover, because transitioning from ePMR to HAMR manufacturing takes meaningfully longer than moving between successive HAMR generations (e.g., Mozaic 3 to Mozaic 4), WDC's HAMR ramp could prove slower than expected, whereas Seagate's head start lets it iterate to higher-capacity HAMR generations more quickly. **Combined with disciplined industry supply and the ASP upcycle, this makes STX a steady long-term compounder, and we reiterate it as the top pick in our coverage.** For additional details, please see [STX FQ4'26: This ain't over... Raising estimates and TP to \\$1350; Reiterate Outperform, Top Long-term pick.](#)

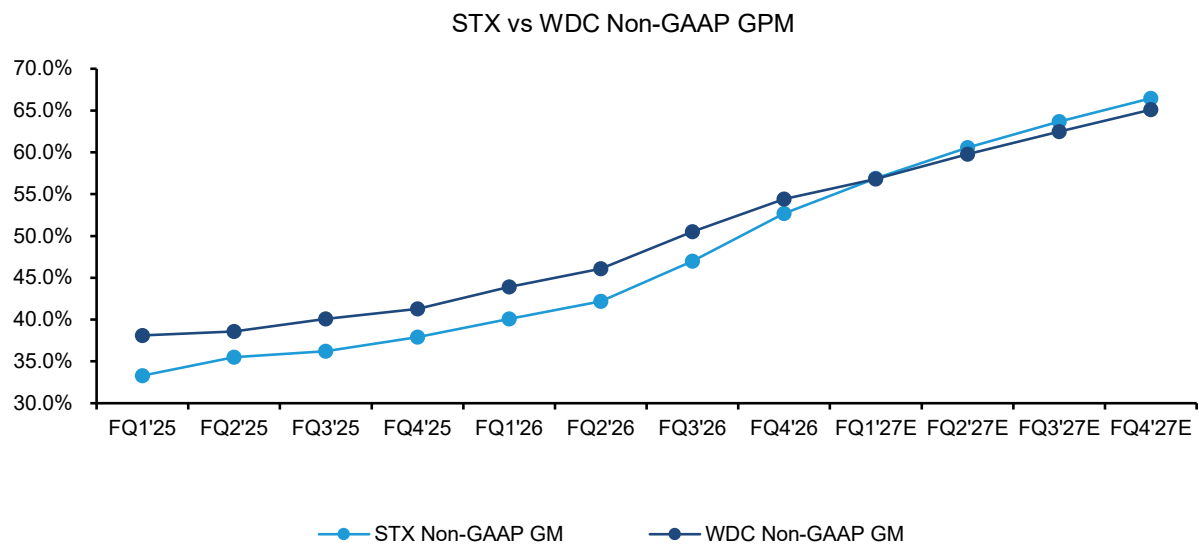
On the other hand, WDC also delivered a strong beat-and-raise quarter, driven largely by favorable pricing, but its guidance falls short of STX's because WDC is lagging on HAMR. That lag constrains WDC's areal density scaling, which in turn limits its exabyte shipment growth and cost-reduction trajectory relative to STX. **A rising tide is lifting all boats, and we remain constructive on both names — but STX's HAMR lead is why it is our top pick.** For additional details on WDC, please see: [WDC FQ4'26: Waiting for HAMR time, but pricing saves the day; Reiterate Outperform, TP raised to \\$770.](#)

EXHIBIT 1: STX shipment YoY Growth above WDC's due to superior areal density scaling


Source: Company reports, Bernstein estimates and analysis

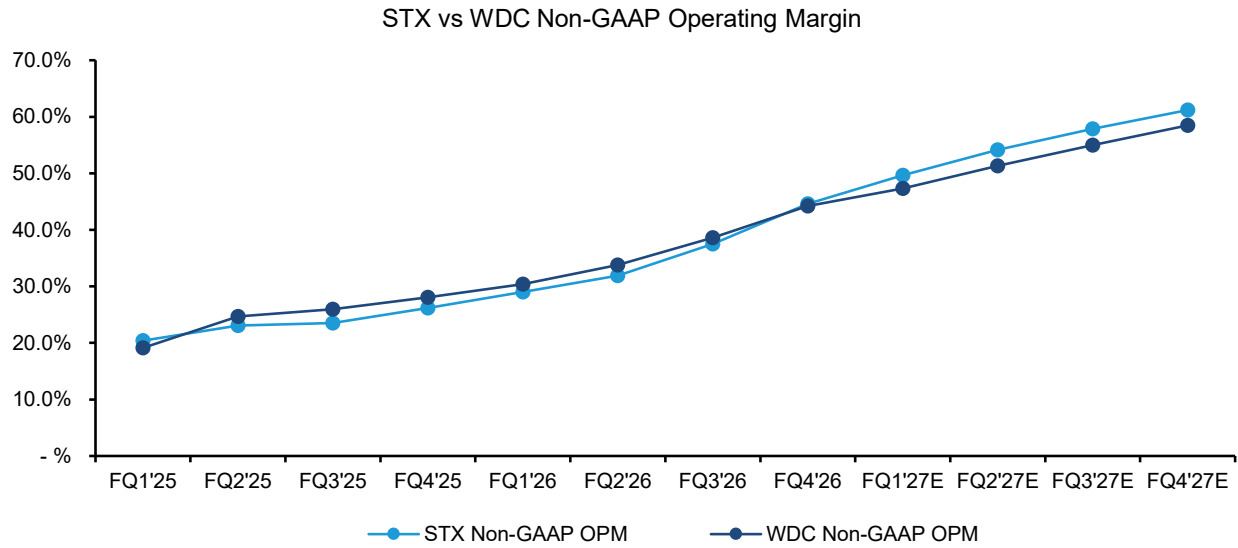
EXHIBIT 2: STX nearline COGS decline is larger than WDC's due to STX's HAMR leadership


Source: Company reports, Bernstein estimates and analysis

EXHIBIT 3: STX will enjoy superior GPM as the leader in HAMR


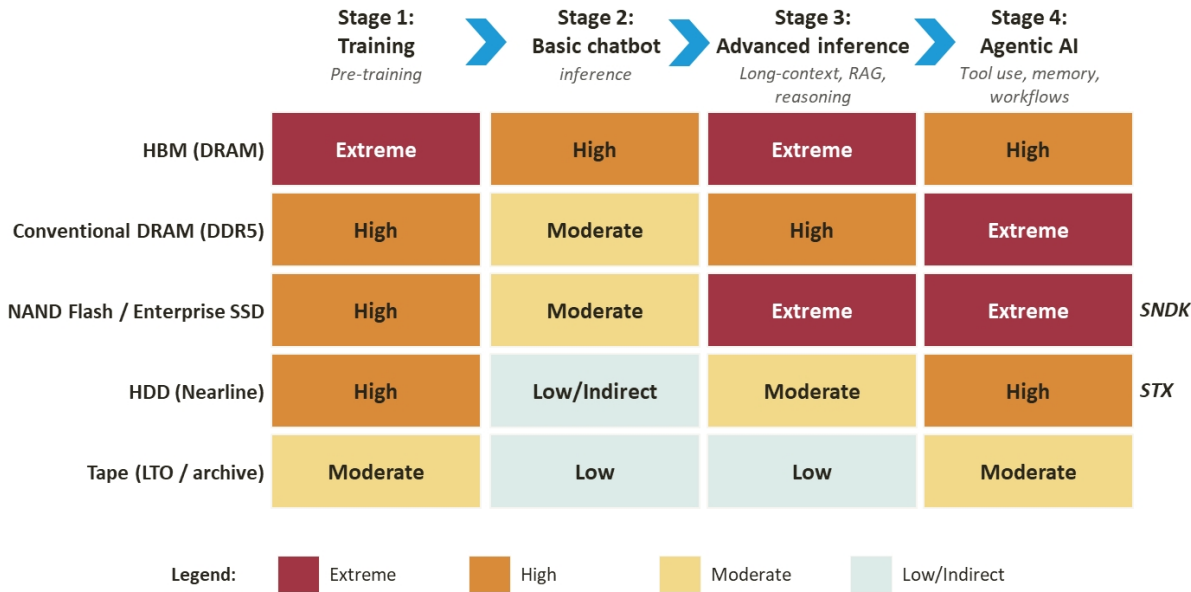
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 4: **STX will enjoy better OPM due to HAMR leadership**



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 5: **The four stages of AI**



Source: Bernstein analysis

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein's "affiliates" relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY**Seagate Technology PLC**

We value Seagate at 21x our FY28 EPS of \$64.40, implying \$1,350 per share. We believe Seagate's improving fundamentals, >70% 5year EPS CAGR, and HAMR leadership easily justifies a 21x P/E, and arguably much higher.

Western Digital Corp

We rate WDC Outperform and have a TP of \$770, on 21x our FY28 EPS.

SanDisk Corp

We value SNDK at \$3000 on 11x FY28 EPS, or 14x through cycle EPS (FY26-30 avg EPS).

RISKS**Seagate Technology PLC**

The biggest risks to the downside on Seagate and to our price target are that: The HDD market is heavily exposed to hyperscale cloud, and could be impacted by a cloud capex digestion or else incremental pressure from changes in hyperscaler buying patterns. 2) WDC could close the gap in HAMR more quickly than expected, shrinking the scope for market share gain and margin improvement. 3) NAND could see better than expected technology improvement and take share from HDDs.

Western Digital Corp

The biggest risks to the downside on WDC and to our price target are that: 1) The HDD market is heavily exposed to hyperscale cloud, and could be impacted by a cloud capex digestion or else incremental pressure from changes in hyperscaler buying patterns; 2) NAND could see better than expected technology improvement and take share from HDDs; and 3) WDC could face significant risks ahead with its HAMR technology transition which could provide a drag on gross margin and EPS.

SanDisk Corp

The biggest risks to the downside on Sandisk and to our price target are that: 1) Near-term numbers look high and face cyclical downside from NAND; 2) SNDK's disclosures and broader investor communications have been confusing, potentially dissuading more quality-focused investors; and 3) NAND weakness could extend beyond the current cycle and be more structural, in which case Sandisk's DCF value could be structurally lower and the impaired value of its assets may be materially below replacement cost.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIA) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- **Outperform (OP):** The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- **Neutral (N):** The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- **Underperform (UP):** The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- **Credit Outperform (C-OP):** The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Neutral (C-N):** The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Underperform (C-UP):** The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

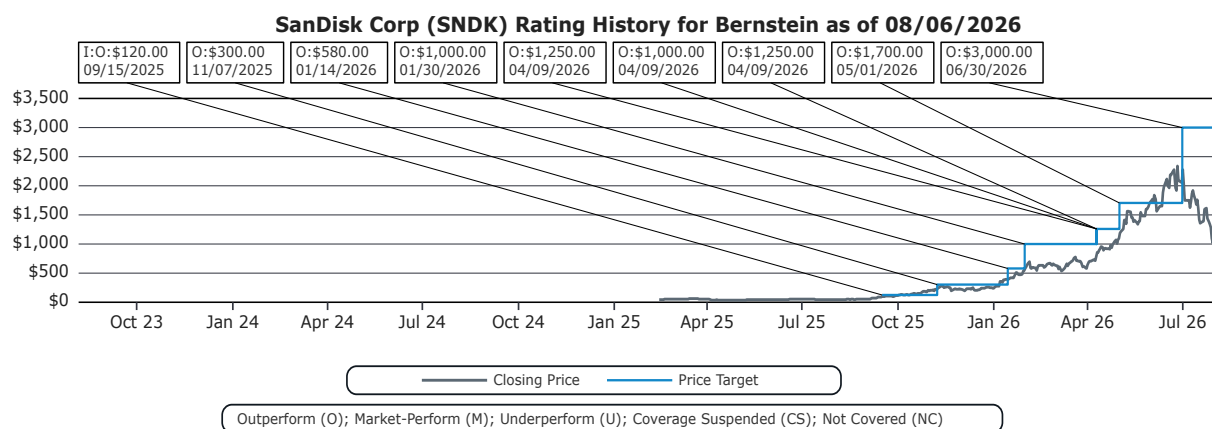
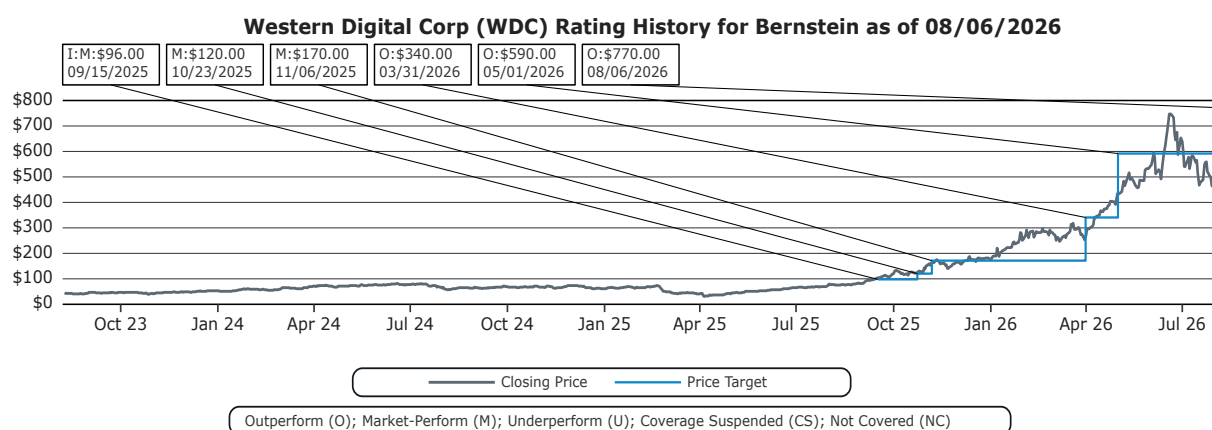
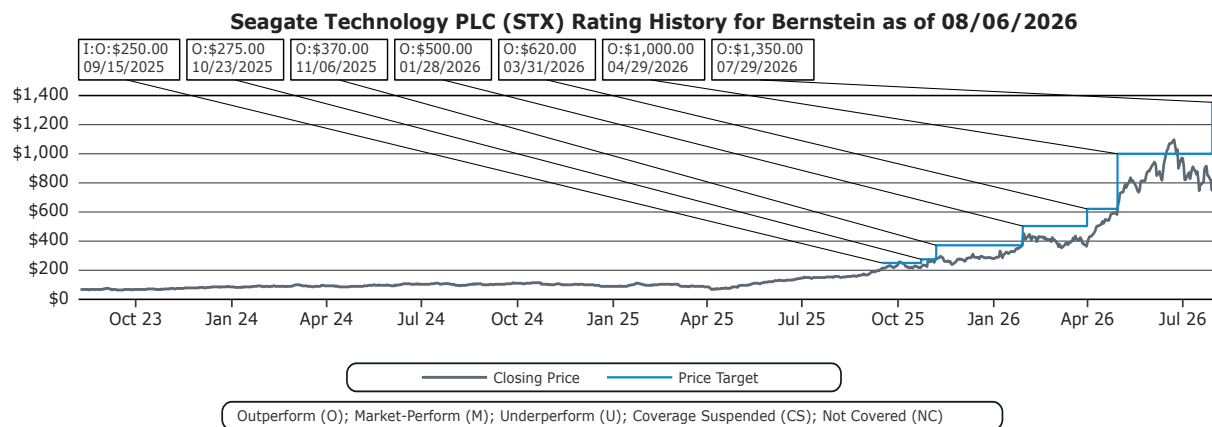
A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side) within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.2%	15.3%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.
As of June 30, 2026. All figures are updated quarterly.

PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY

All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

CONFLICTS OF INTEREST

An affiliate of Bernstein has received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Western Digital Corp.

OTHER MATTERS

The legal entity(ies) employing the analyst(s) listed in this report, and their location, can be determined by the country code of their phone number, as follows:

- +1 Bernstein Institutional Services LLC; New York, New York, USA
- +44 Bernstein Autonomous LLP; London UK
- +212 Société Générale Africa Technologies & Services; Casablanca, Morocco
- +33 BSG France S.A.; Paris, France
- +34 BSG France S.A.; Madrid, Spain
- +41 Bernstein Autonomous LLP; Geneva, Switzerland
- +49 BSG France S.A.; Frankfurt, Germany
- +91 Sanford C. Bernstein (India) Private Limited; Mumbai, India
- +852 Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司; Hong Kong, China
- +65 Sanford C. Bernstein (Singapore) Private Limited; Singapore
- +81 Sanford C. Bernstein Japan KK; Tokyo, Japan

Where this report has been prepared by research analyst(s) employed by a non-US affiliate, such analyst(s), is/are (unless otherwise expressly noted below) not registered as associated persons of Bernstein Institutional Services LLC or any other SEC-registered broker-dealer and are not licensed or qualified as research analysts with FINRA. Accordingly, such analyst(s) may not be subject to FINRA's restrictions regarding (among other things) communications by research analysts with a subject company, interactions between research analysts and investment banking personnel, participation by research analysts in solicitation and marketing activities relating to investment banking transactions, public appearances by research analysts, and trading securities held by a research analyst account.

Where this report has been prepared by research analyst(s) employed by Société Générale Africa Technologies & Services (part of the Société Générale group of companies), it has been prepared on behalf of a Bernstein company under a Global Services Agreement in place between Bernstein and Société Générale.

CERTIFICATION

Each research analyst listed in this report, who is primarily responsible for the preparation of the content of this report, certifies that all of the views expressed in this publication accurately reflect that analyst's personal views about any and all of the subject securities or issuers and that no part of that analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views in this publication.

II. ADDITIONAL GLOBAL CONFLICT DISCLOSURES

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e., the private side) within the Firm, and into other areas, units, groups or affiliates (i.e., public side) of the Firm.

III. OTHER IMPORTANT INFORMATION AND DISCLOSURES

Separate branding is maintained for "Bernstein" and "Autonomous" research products.

- Bernstein produces a number of different types of research products including, among others, fundamental analysis and quantitative analysis under both the "Autonomous" and "Bernstein" brands. Recommendations contained within one type of research product may differ from recommendations contained within other types of research products, whether as a result of differing time horizons, methodologies or otherwise. Furthermore, views or recommendations within a research product issued under one brand may differ from views or recommendations under the same type of research product issued under the other brand. The Research Ratings System for the two brands and other information related to those Rating Systems are included in the previous section.

- Autonomous operates as a separate business unit within the following entities: Bernstein Institutional Services LLC, Bernstein Autonomous LLP, Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 and Sanford C. Bernstein (India) Private Limited. For information relating to “Autonomous” branded products (including certain Sales materials) please visit: www.autonomous.com. For information relating to Bernstein branded products please visit: www.bernsteinresearch.com.

Analysts are compensated based on aggregate contributions to the research franchise as measured by account penetration, productivity and proactivity of investment ideas. No analysts are compensated based on performance in, or contributions to, generating investment banking revenues.

This report has been produced by an independent analyst as defined in Article 3 (1)(34)(i) of EU 596/2014 Market Abuse Regulation (“MAR”) and the same article of MAR as it forms part of United Kingdom domestic law by virtue of the European Union (Withdrawal) Act 2018.

To our readers in the United States: Bernstein Institutional Services LLC, a broker-dealer registered with the U.S. Securities and Exchange Commission (“SEC”) and a member of the U.S. Financial Industry Regulatory Authority, Inc. (“FINRA”) is distributing this publication in the United States and accepts responsibility for its contents. Where this material contains an analysis of debt product(s), such material is intended only for institutional investors and is not subject to the US independence and disclosure standards applicable to debt research prepared for retail investors.

Bernstein Institutional Services LLC may act as principal for its own account or as agent for another person (including an affiliate) in sales or purchases of any security which is a subject of this report. This report does not purport to meet the objectives or needs of any specific individuals, entities or accounts.

To our readers in Canada: If this publication pertains to a Canadian domiciled company, it is being distributed in Canada by Sanford C. Bernstein (Canada) Limited, which is licensed and regulated by the Canadian Investment Regulatory Organization. If the publication pertains to a non-Canadian domiciled company, it is being distributed by Bernstein Institutional Services LLC, which is licensed and regulated by both the SEC and FINRA, into Canada under the International Dealers Exemption.

This document may not be passed onto any person in Canada unless that person qualifies as “permitted client” as defined in Section 1.1 of NI 31-103.

To our readers in Brazil: This report has been prepared by Bernstein Institutional Services LLC, and Banco BTG Pactual S.A. (“BTG”) is responsible for the distribution of this report in Brazil.

To readers in the United Kingdom: This publication has been issued or approved for issue in the United Kingdom by Bernstein Autonomous LLP, authorised and regulated by the Financial Conduct Authority and located at 60 London Wall, London EC2M 5SH, +44 (0)20-7170-5000. Registered in England & Wales No OC343985.

This document is for distribution only to persons who (i) have professional experience in matters relating to investments falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the “Financial Promotion Order”), (ii) are persons falling within Article 49(2)(a) to (d) (“high net worth companies, unincorporated associations, etc.”) of the Financial Promotion Order, (iii) are outside the United Kingdom, or (iv) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the FSMA) in connection with the issue or sale of any securities may otherwise lawfully be communicated or caused to be communicated (all such persons together being referred to as “relevant persons”). This document is directed only at relevant persons and must not be acted on or relied on by persons who are not relevant persons. Any investment or investment activity to which this document relates is available only to relevant persons and will be engaged in only with relevant persons.

To our readers in the member states of the EEA: This publication is being distributed by BSG France SA, which is authorised and regulated by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and Autorité des Marchés Financiers (AMF).

To our readers in Hong Kong: This publication is being distributed in Hong Kong by Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, which is licensed and regulated by the Hong Kong Securities and Futures Commission (Central Entity No. AXC846) to carry out Type 4 (Advising on Securities) regulated activities and subject to the licensing conditions mentioned in the SFC Public Register (<https://www.sfc.hk/publicregWeb/corp/AXC846/details>). This publication is solely for professional investors, as defined in the Securities and Futures Ordinance (Cap. 571). The purpose of this report is solely to provide an analysis of the issuers referred to in this report and is not intended for any purpose contrary to the laws of Hong Kong.

To our readers in Singapore: This publication is being distributed in Singapore by Sanford C. Bernstein (Singapore) Private Limited, only to accredited investors or institutional investors, as defined in the Securities and Futures Act 2001 of Singapore (“SFA”). Recipients in Singapore should contact Sanford C. Bernstein (Singapore) Private Limited in respect of matters arising from, or in connection with, this publication. Sanford C. Bernstein (Singapore) Private Limited is regulated by the Monetary Authority of Singapore and licensed under the SFA as a capital markets services licence holder for dealing in capital markets products that are

securities and collective investment schemes and an exempt financial adviser for advising on, issuing and promulgating analyses and reports on securities. Sanford C. Bernstein (Singapore) Private Limited is registered in Singapore with Company Registration No. 20213710W and located at 8 Marina Boulevard, #12-01, Marina Bay Financial Centre, Singapore 018981, +65-6326-7000.

To our readers in the People's Republic of China: The securities referred to in this document are not being offered or sold and may not be offered or sold, directly or indirectly, in the People's Republic of China (for such purposes, not including the Hong Kong and Macau Special Administrative Regions or Taiwan, the "PRC") in contravention of any applicable laws of the PRC.

This document does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC to any person to whom it is unlawful to make the offer or solicitation in the PRC.

We do not represent that this document may be lawfully distributed, or that any securities may be lawfully offered, in compliance with any applicable registration or other requirements in the PRC, or pursuant to an exemption available thereunder, or assume any responsibility for facilitating any such distribution or offering. In particular, no action has been taken by us which would permit a public offering of any securities or distribution of this document in the PRC. Accordingly, the securities are not being offered or sold within the PRC by means of this document or any other document. Neither this document nor any advertisement or other offering material may be distributed or published in the PRC, except under circumstances that will result in compliance with any applicable laws and regulations.

To our readers in Japan: This publication is being distributed in Japan by Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社), which is registered in Japan as a Financial Instruments Business Operator with the Kanto Local Finance Bureau (registration number: The Director-General of Kanto Local Finance Bureau (FIBO) No.3387) and regulated by the Financial Services Agency. It is also a member of Investment Management Association of Japan. This publication is solely for qualified institutional investors in Japan only, as defined in Article 2, paragraph (3), items (i) of the Financial Instruments and Exchange Act.

For the institutional client readers in Japan who have been granted access to the Bernstein website by Daiwa Securities Group Inc. ("Daiwa"), your access to this document should not be construed as meaning that Bernstein is providing you with investment advice for any purposes. Whilst Bernstein has prepared this document, your relationship is, and will remain with, Daiwa, and Bernstein has neither any contractual relationship with you nor any obligations towards you.

To our readers in Australia: Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is responsible for distributing research in Australia. It is regulated by the Securities and Exchange Commission under U.S. laws, by the Financial Conduct Authority under U.K. laws, which differs from Australian laws. Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is exempt from the requirement to hold an Australian financial services license under the Corporations Act 2001 in respect of the provision of the following financial services to wholesale clients:

- providing financial product advice;
- dealing in a financial product;
- making a market for a financial product; and
- providing a custodial or depository service.

To our readers in India: This publication is being distributed in India by Sanford C. Bernstein (India) Private Limited (SCB India) which is licensed and regulated by Securities and Exchange Board of India ("SEBI") as a research analyst entity under the SEBI (Research Analyst) Regulations, 2014, having registration no. INH000006378 and as a stock broker having registration no. INZ000213537. SCB India is currently engaged in the business of providing research and stock broking services. Please refer to www.bernsteinresearch.in for more information.

- SCB India is a Private limited company incorporated under the Companies Act, 2013, on April 12, 2017 bearing corporate identification number U65999MH2017FTC293762, and registered office at Level 3A, 4th Floor, First International Financial Centre, Plot Nos C-54 and C-55, G Block, Near CBI Office, Bandra Kurla Complex, Bandra (East), Mumbai 400098, Maharashtra, India (Phone No: +91-22-68421401).
- For details of Associates (i.e., affiliates/group companies) of SCB India, kindly email MUM-BERNSTEIN-InCompliance@bernsteinsg.com.
- SCB India does not have any disciplinary history as on the date of this report.

- Except as noted above, SCB India and/or its Associates (i.e., affiliates/group companies), the Research Analysts authoring this report, and their relatives
 - do not have any financial interest in the subject company
 - do not have actual/beneficial ownership of one percent or more in securities of the subject company;
 - is not engaged in any investment banking activities for Indian companies, as such;
 - have not managed or co-managed a public offering in the past twelve months for any Indian companies;
 - have not received any compensation for investment banking services or merchant banking services from the subject company in the past 12 months;
 - have not received compensation for brokerage services from the subject company in the past twelve months;
 - have not received any compensation or other benefits from the subject company or third party related to the specific recommendations or views in this report; and
 - do not currently, but may in the future, act as a market maker in the financial instruments of the companies covered in the report.
 - do not have any conflict of interest in the subject company as of the date of this report.
- Except as noted above, the subject company has not been a client of SCB India during twelve months preceding the date of distribution of this research report. Neither SCB India nor its Associates (i.e., affiliates/group companies) have received compensation for products or services other than investment banking, merchant banking or brokerage services from the subject company in the past twelve months.
- The principal research analyst(s) who prepared this report, members of the analysts' team, and members of their households are not an officer, director, employee or advisory board member of the companies covered in the report.
- Our Compliance officer / Grievance officer is Ms. Rupal Talati, who can be reached at +91-22-68421451, or MUM-BERNSTEIN-InCompliance@bernsteinsg.com / Scbin-investorgrievance@bernsteinsg.com
- The Research investor charter and Terms & Conditions of SCB India are available on its website and may be accessed at [Sanford C. Bernstein \(India\) Private Limited](https://bernsteinresearch.in/) (<https://bernsteinresearch.in/>) for your reference.
- Disclaimer: Registration granted by SEBI, and certification from NISM, is in no way a guarantee of performance of the intermediary or provide any assurance of returns to investors. Investments in securities market are subject to market risks. Read all the related documents carefully before investing.

To our readers in Switzerland: This document is provided in Switzerland by or through Bernstein Autonomous LLP, and is provided only to qualified investors as defined in article 10 of the Swiss Collective Investment Scheme Act ("CISA") and related provisions of the Collective Investment Scheme Ordinance and in strict compliance with applicable Swiss law and regulations. The products mentioned in this document may not be suitable for all types of investors. This document is based on the Directives on the Independence of Financial Research issued by the Swiss Bankers Association (SBA) in January 2008.

To our readers in the Middle East: Bernstein Autonomous LLP, DIFC branch has its principal office at Gate Village 06, DIFC, Dubai, UAE. Bernstein Autonomous LLP, DIFC branch is regulated by the Dubai Financial Services Authority (DFSA) with the registration number CL10040 and is provisioned for Arranging Deals in Investments and Advising on Financial Products. All communications and services are directed at Professional Clients and Market Counterparties only (as defined in the DFSA rulebook). Persons other than Professional Clients and Market Counterparties, such as Retail Clients, are not the intended recipients of our communications or services.

LEGAL

All research publications are disseminated to our clients through posting on the firm's password protected websites, bernsteinresearch.com and autonomous.com. Certain, but not all, research publications are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience.

This publication has been published and distributed in accordance with the Firm's policy for management of conflicts of interest in investment research, a copy of which is available from Bernstein Institutional Services LLC, Director of Compliance, 245 Park Avenue, New York, NY 10167. Additional disclosures and information regarding Bernstein's business are available on our website www.bernsteinresearch.com.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. This publication is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of, or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or which would subject any of the entities referenced herein or any of their subsidiaries or affiliates to any registration or licensing requirement within such jurisdiction. This publication is based upon public sources we believe to be reliable, but no representation is made by us that the publication is accurate or complete. We do not undertake to advise you of any change in the reported information or in the opinions herein. This publication was prepared and issued by entity referred to herein for distribution to eligible counterparties or professional clients. This publication is not an offer to buy or sell any security, and it does not constitute investment, legal or tax advice. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with their professional advisors in light of their specific circumstances. The value of investments may fluctuate, and investments that are denominated in foreign currencies may fluctuate in value as a result of exposure to exchange rate movements. Information about past performance of an investment is not necessarily a guide to, indicator of, or assurance of, future performance.

This report is directed to and intended only for our clients who are "eligible counterparties", "professional clients", "institutional investors" and/or "professional investors" as defined by the aforementioned regulators, and must not be redistributed to retail clients as defined by the aforementioned regulators. Retail clients who receive this report should note that the services of the entities noted herein are not available to them and should not rely on the material herein to make an investment decision. The result of such act will not hold the entities noted herein liable for any loss thus incurred as the entities noted herein are not registered/ authorised/ licensed to deal with retail clients and will not enter into any contractual agreement/arrangement with retail clients. This report is provided subject to the terms and conditions of any agreement that the clients may have entered into with the entities noted herein. All research reports are disseminated on a simultaneous basis to eligible clients through electronic publication to our client portal.

The information in this report was prepared by Bernstein solely for the internal business use of our clients. Clients may store, display, analyze, reformat and print the information in this report for this limited use only. Clients may not copy, alter, create derivative works, resell, reverse engineer, commercially exploit, share or distribute any part of the information contained herein for any purpose without Bernstein's express written consent. These restrictions include extracting data or using the content to develop indices or other products. Further, you may not use this report, or any portion of this report, to train or finetune any third-party machine learning or artificial intelligence system, or as a prompt or input into any such system. You also may not, without Bernstein's express written consent, do any of the foregoing in connection with your own internal machine learning or artificial intelligence system.

Bernstein may use artificial intelligence tools in the preparation of its materials. Any such materials are reviewed by Bernstein's research analysts prior to publication.

This report has been prepared for information purposes only and is based on current public information that we consider reliable, but the entities noted herein do not warrant or represent (express or implied) as to the sources of information or data contained herein are accurate, complete, not misleading or as to its fitness for the purpose intended even though the entities noted herein rely on reputable or trustworthy data providers, it should not be relied upon as such. Opinions expressed are the author(s)' current opinions as of the date appearing on the material only and we do not undertake to advise you of any change in the reported information or in the opinions herein.

Any references to Societe Generale herein are purely factual, based upon publicly available information, and included for comparative purposes only. They do not constitute an opinion or recommendation with respect to the securities of Societe Generale.

This publication was prepared and issued by the entity referred to herein for distribution to eligible counterparties or professional clients. The information in this report is intended for general circulation and does not constitute an offer to buy or sell any security, investment, legal or tax advice nor a personal recommendation, as defined by any of the aforementioned regulators. It does not take into account the particular investment objectives, financial situations, or needs of individual investors. The report has not been reviewed by any of the aforementioned regulators and does not represent any official recommendation from the aforementioned regulators. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with advice sought from a financial adviser regarding the suitability of the investment product, taking into account the specific investment objectives, financial situation or particular needs of any recipient of the recommendation, before the recipient makes a commitment to purchase the investment product.

The analysis contained herein is based on numerous assumptions. Different assumptions could result in materially different results. The information in this report does not constitute, or form part of, any offer to sell or issue, or any offer to purchase or subscribe for shares, or to induce engagement in any other investment activity. The value of any securities or financial instruments mentioned in this report may fluctuate subject to market conditions. Information about past performance of an investment is not necessarily a guide to, indicative of, or assurance of future performance. Estimates of future performance mentioned by the research analyst in this report are based on assumptions that may not be realized due to unforeseen factors like market volatility/fluctuation. In relation to securities or financial instruments denominated in a foreign currency other than the clients' home currency, movements in exchange rates will have an effect on the value, either favorable or unfavorable. Before acting on any recommendations in this report, recipients should consider the appropriateness of investing in the subject securities or financial instruments mentioned in this report and, if necessary, seek independent professional advice.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject the entities noted herein to any regulation or licensing requirement within such jurisdiction.

Source: Bloomberg Index Services Limited. BLOOMBERG® is a trademark and service mark of Bloomberg Finance L.P. and its affiliates (collectively "Bloomberg"). Bloomberg or Bloomberg's licensors own all proprietary rights in the Bloomberg Indices. Neither Bloomberg nor Bloomberg's licensors approves or endorses this material, or guarantees the accuracy or completeness of any information herein, or makes any warranty, express or implied, as to the results to be obtained therefrom and, to the maximum extent allowed by law, neither shall have any liability or responsibility for injury or damages arising in connection therewith.

No part of this material may be reproduced, distributed or transmitted or otherwise made available without prior consent of the entities noted herein. Copyright Bernstein Institutional Services LLC Bernstein Autonomous LLP, BSG France S.A., Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited and Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社). All rights reserved. The trademarks and service marks contained herein are the property of their respective owners. Any unauthorized use or disclosure is strictly prohibited. The entities noted herein may pursue legal action if the unauthorized use results in any defamation and/or reputational risk to the entities noted herein and research published under the Bernstein and Autonomous brands.